

Edge-Based Defense in Networks

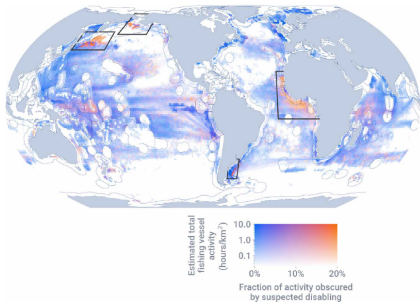
A Modification of Bloch, Chatterjee & Dutta (2023)

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Estimated global fishing activity. Orange marks suspected illegal activity — West Africa is one of three concern zones.

Welch et al., *Science Advances* (2022) • Global Fishing Watch

The problem.

Illegal fishing reaches places no single country owns. Vessels move along sea routes between national waters and high seas, taking what isn't theirs to take.

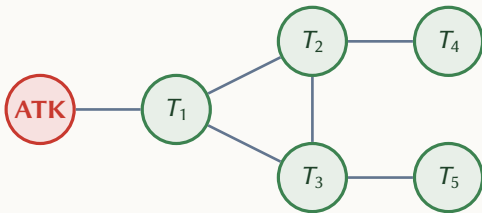
The adversary is strategic.

Routes are chosen, not random. The weakest gap is the one used.

The defenders are many.

Neighbouring states watch the same waters. Defense is naturally shared.

A network is a graph $G = (V, E)$: nodes V are locations, edges E are direct connections between them.

**Nodes.**

Locations the attacker can reach.

Edges.

The connections between locations.

The attacker.

A single adversary outside the network.

The defenders.

One at each target. They invest to intercept.



	Node Defense		Edge Defense
	Centralized	Continuous	Centralized
<i>Authors</i>	Dziubiński & Goyal Cerdeiro et al.	Bloch, Chatterjee & Dutta (2023)	Washburn & Wood Mavronicolas et al.
<i>Investment</i>	Binary	Continuous	Discrete or budgeted
<i>Defenders</i>	Single planner	Many	Single planner

The gap — *Continuous, decentralized investment — placed on the connections.*

$$\alpha_{k(p)}(p) = \prod_{\substack{j \in V(p) \\ j \neq 0, j \neq k(p)}} (1 - x_j)$$

$$U(\pi, x) = \sum_{p \in P} \pi(p) \beta_{k(p)}(p) b_{k(p)}$$

$$V_i(\pi, x) = - \sum_{\substack{p \in P \\ k(p)=i}} \pi(p) \beta_i(p) b_i - \frac{x_i^2}{2}$$

Path survival.

The attacker survives the interior of the path if it slips past every defender between the attacker node and the target.

Attacker's expected payoff.

Sum over every path, weighted by how often the attacker uses it, times the chance of getting through, times the target's value.

Defender's payoff.

Expected loss from attacks on its own target, minus the cost of its investment. Defenders are the only players who pay a cost.

Bloch distinguishes two kinds of target. A first target is reached directly from the attacker node; a deeper target is reached only by passing through other targets that are themselves attacked.

$$x_i = 1 - \frac{U}{b_i}, \quad q_i = \frac{1}{b_i} \left(1 - \frac{U}{b_i} \right)$$

$$x_i = 1 - \frac{b_k}{b_i}, \quad q_i = \frac{b_k}{b_i U} \left(1 - \frac{b_k}{b_i} \right)$$

First-target equilibrium.

Higher-value targets invest more and are attacked more often. U is the common payoff across all attacked targets.

Deeper-target equilibrium.

For a target i reached only through its predecessor k . The predecessor's value b_k replaces U as the relevant threshold.

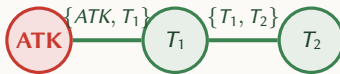
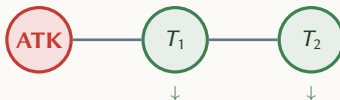
Defense lives at the location.

In Bloch, every defender's investment sits at its own node. The attacker is caught *at* a target, not on the way to one.

But corridors are shared.

A shipping lane, a forest road, a fishing zone — none of these belong to one party alone. Real defense happens *on the way*, by whoever watches that stretch.

Bloch: defense at nodes



Edge-based: defense on edges

Move investment from nodes to edges. Every edge has two defenders.

Bloch's baseline

$$S(\text{node } i) = 1 - x_i$$

One defender per node. Survival is whatever it fails to catch.

The modification

$$S(e) = (1 - x_{a,e})(1 - x_{b,e})$$

Two defenders per edge. The attacker escapes only if *both* sides fail.

Cost is split: each defender i pays $\sum_{e \ni i} x_{i,e}^2 / 2$, summed over its incident edges.

Three of the four key lemmas in Bloch's analysis depend only on the attacker's optimization. They survive the move to edges.

Equal payoff across the support

Every attacked target gives the attacker the same payoff U .

Optimal predecessor has lowest value

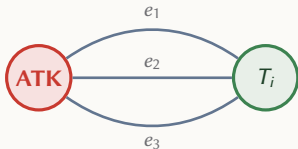
If a target can be reached through several support members, the attacker passes through the one with the lowest value.

Values increase along attack paths

Target values rise as the attacker moves deeper. Survival decay is offset by rising value.

Plus: equilibrium exists and is unique, under the same genericity assumption Bloch uses.

A target reachable through several free-highway edges is structurally weaker: each edge must be defended separately.



$m_i = 3$ free-highway edges
reaching the same target

Bloch: $q_i = \frac{b_i - U}{b_i^2}$

Edge-based: $q_i = \frac{m_i (b_i - U)}{b_i^2}$

The factor m_i counts the free-highway edges feeding the target.
More edges $\Rightarrow$ more attack probability $\Rightarrow$ weaker target.

When two defenders share an edge, one's investment changes the other's incentive.

$$x_{i,e}^* = q b_i (1 - x_{j,e})$$

$$\frac{\partial x_{i,e}^*}{\partial x_{j,e}} = -q b_i < 0$$

If my neighbor invests more, I invest less.

The slope is strictly negative whenever attacks flow through the edge. Defenders' efforts substitute for each other.

But the edge cannot stay undefended.

If the other side invests zero, my best response is strictly positive. $(0, 0)$ is never an equilibrium when attacks reach the edge.

Three further differences from Bloch's framework, briefly.

Mutual coverage on edges

When a target has m_i free-highway edges, the attacker spreads attack equally across them, and the defender invests equally on each.

No Braess paradox

Adding a connection in Bloch can hurt the attacker. Here, every new free-highway edge strictly raises attacker payoff U . Braess has no room to operate.

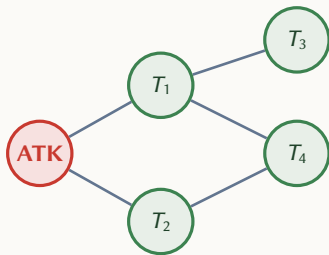
Mixed configurations

A target can be reached by both free-highway edges *and* through a predecessor. Each edge gets its own equilibrium equation. No analog in Bloch.

Two endpoints, two modelling choices. Crossing them gives four cases.

	Independent interception	Joint interception
	$S(e) = (1 - x_{i,e})(1 - x_{j,e})$	$S(e) = 1 - x_{i,e} x_{j,e}$
Separable cost $x_{i,e}^2 / 2$ each	Case A (this chapter) Substitutes. Edge never undefended.	Case C Complements. (0, 0) possible.
Shared cost $(x_i + x_j)^2 / 4$ each	Case B Stronger substitutes. Edge never undefended.	Case D Threshold at $q b_i = 1/2$.

Headline: technology matters more than cost — it decides whether an undefended edge can persist.



Values: $b_1=0.20$, $b_2=0.25$, $b_3=0.50$, $b_4=0.70$

T_4 has $m_4 = 2$ (reached via T_1 and T_2)

	Bloch	Edge-based
Attacker utility U	0.4020	0.4773 (+18.7%)
Attack on T_4 (q_4)	0.6081	0.9091
Attack on T_3 (q_3)	0.3920	0.0909
Investment x_4	0.4257	0.3181 per edge
Investment x_3	0.1960	0.0455

More edges to defend $\Rightarrow$ higher $U \Rightarrow T_3$ pushed to the margin $\Rightarrow$ attacks concentrate on T_4 .

A single coordinator chooses all investments to minimize total expected loss across the network.

$$L = \sum_{i \in \Delta} \sum_{p \in P_i} \pi(p) \left[\prod_{e \in p} (1 - y_{a_e, e})(1 - y_{b_e, e}) \right] b_i + \sum_{e \in E} \sum_{j \in e} \frac{y_{j, e}^2}{2}$$

Symmetry within edges.

The two endpoints of any shared edge invest the same amount. The coordinator treats them identically.

The two terms.

Expected loss across every attacked target and every path, plus total cost paid by both endpoints of every edge.

Same harm, lower cost (on lines).

Convexity makes splitting cheaper than concentration. The cost savings show up on line networks.



Centralized defender. T_2 is dropped from the support and used as a shield in both formulations.

	Node defense	Edge-based
Attacker utility U	0.2400	0.2400
Total defense cost	0.1396	0.1142
Cost saving	—	18.2%

Same attacker harm, lower cost. Splitting interception between two payers costs less than concentrating it on one. 16 / 24

Real players rarely observe everything. The edge-based model extends in two natural directions.

Private target values.

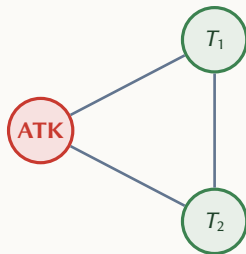
Each defender knows its own value b_i , drawn from a common prior F . The attacker operates under the prior alone.

Partial network observation.

Each player observes its own incident edges with certainty, and assigns probability θ to every unseen edge.

Motivated by cybersecurity: Chang et al. (2024) show attackers target firms by observable signals (size, sales) — not by the true value at stake. The two extensions map onto these two information asymmetries.

Equilibrium concept: Bayes-Nash. Worked examples: the triangle network.



Prior: F uniform on $(0.2, 0.8)$

$$\mu = 0.5.$$

	Complete	Incomplete
Attacker utility U	0.3869	0.3726
Shared edge defended by	T_2 only	both
x on $\{T_1, T_2\}$ by T_1	0.0000	0.0168
x on $\{T_1, T_2\}$ by T_2	0.0392	0.0175

Headline: uncertainty disrupts perfect coordination. Both defenders cover the shared edge — neither knows for sure they are the lower-value target. Attacker is worse off.

Same triangle. Now b_1, b_2 are common knowledge but each player believes any unseen edge exists with probability θ .

θ	Regime
$\in (0, 0.961)$	Direct-only ($\gamma_1 = \gamma_2 = 0$)
$\in [0.961, 1]$	Partial roundabout ($\gamma_2 > 0$)

Threshold: $\theta^* \approx 0.961$

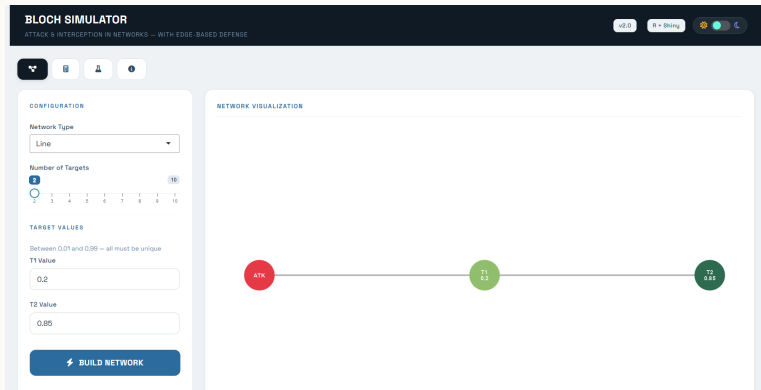
Below the threshold.

The shared edge isn't used by the attacker, so defenders leave it undefended. U is locked at 0.3748 — independent of how uncertain the network is.

Above the threshold.

Roundabout routes become attractive. Defender 2 starts covering the shared edge. As $\theta \rightarrow 1$, U rises smoothly to the complete-info value 0.3869.

Headline: a sharp threshold, not a smooth slope. Below it, uncertainty is “absorbing” — nothing changes.

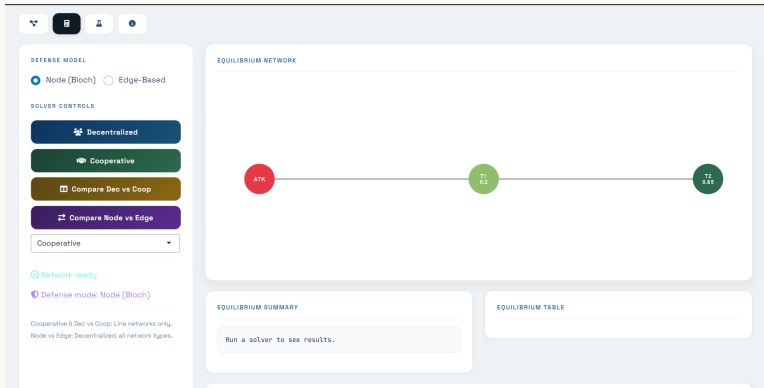


Build any network.

Choose topology, set targets, assign values.

Built-in checks.

Unique values, reachable targets, no orphans.



Two defense models.

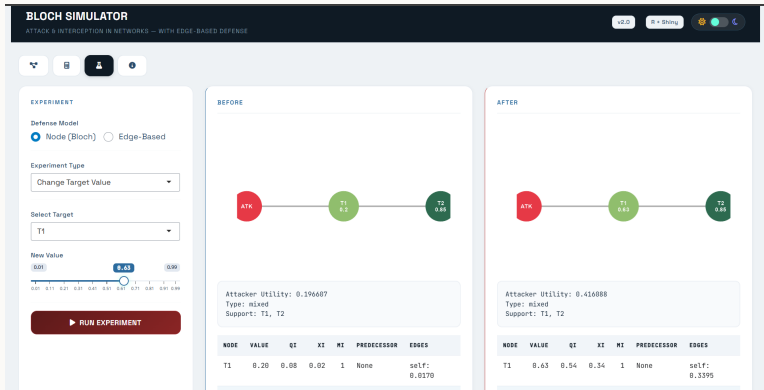
Bloch (nodes) or edge-based.

Two solver modes.

Decentralized or cooperative.

Support reduction.

Automatic, iterative.



Perturb and see.

Change a value, add a connection, or remove a target.

Before and after.

Side-by-side comparison with impact summary.

What this work does not resolve.

Cooperative theory limited

Closed-form cost-savings result holds on line networks and two-branch trees. Networks with multi-path interior targets need separate treatment.

Single attacker only

No multi-adversary version yet. The equilibrium and cooperative analysis would both need extension.

Incomplete info on triangle only

Bayes-Nash results derived on the smallest network where the mechanism appears. Larger networks have more routes and probably more thresholds.

Interior stability open

Strategic substitutes on shared edges raise dynamic questions about how endpoints arrive at the equilibrium. Not addressed here.

Future directions.

- Cooperative defense on deeper trees and multi-path networks
- Joint-cost variation analyzed in full
- Combined value + network uncertainty
- Multi-attacker extension
- Empirical calibration: illegal fishing, logging, waste
- Network design: where should new connections go?

Key takeaways.

- Moving defense from *nodes* to *edges* produces a coherent, decentralized model — a gap that node-defense and edge-defense traditions had each left open.
- Multi-path weakness, strategic substitutes, no Braess paradox, and cooperative cost savings on lines all follow from one structural change.
- The framework survives incomplete information in two distinct settings.

Thank you.

Questions and comments very welcome.